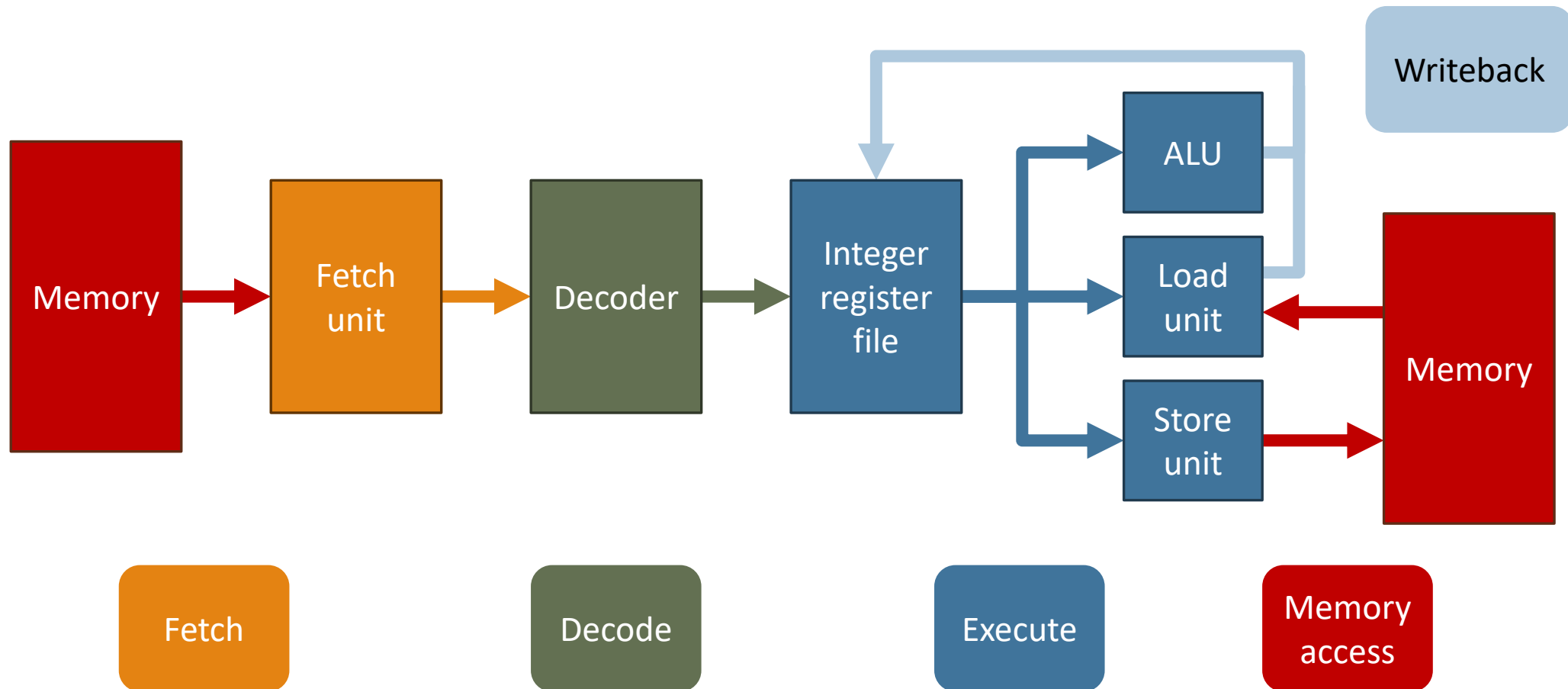


Simple pipelined scalar processor

BY MIŁOSZ WASACZ

Microarchitecture



Features & ISA

Features:

- 5-stage pipeline
 - Fetch, Decode, Execute, Memory Access, Writeback
 - No bypassing
 - Stalling for data hazard prevention
- ELF file parsing (using [elf](#) library)
 - Programs written in C can be compiled using GCC from the included Docker image and run on the simulator (provided they use only supported system calls)

ISA:

- [RISC-V unprivileged ISA](#)
 - Support for all instructions in RV32I Base Integer ISA (except for FENCE and EBREAK)
 - Support for a selection of system calls
 - exit, close, lseek, read, sbrk, write



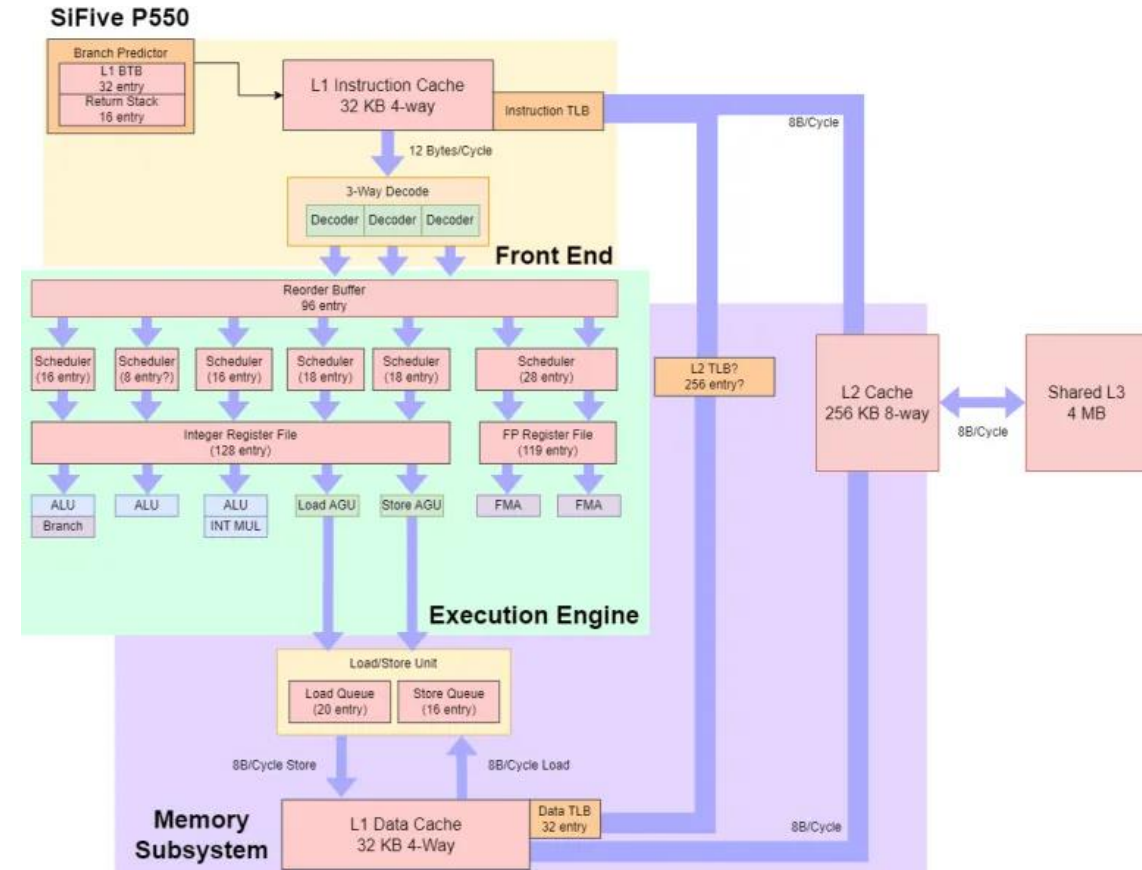
Source: <https://riscv.org/>

Future Microarchitecture & Features

The target microarchitecture will be based SiFive P550, potentially with the addition of vector/SIMD operations.

Planned features (in order of implementation):

- Bypassing
- Reorder buffer, schedulers & register renaming
- Superscalar execution (more decoders)
- Branch prediction
- Support for RV64I ISA & “M”, “F”, “D” Extensions
- Vector/SIMD instructions (“V” Extension)
- Memory hierarchy



Source: <https://chipsandcheese.com/p/inside-sifives-p550-microarchitecture>

Planned benchmarks & experiments

Experiments:

- Pipelining & superscalar execution effectiveness
- Branch prediction effectiveness
- Vector operations vs no vector operations
- Cache policy effectiveness

Benchmarks:

- Sorting – bubblesort, quicksort, mergesort
- GCD
- IMATMUL
- Livermore Loops
- Recursive algorithms – factorial, binary search